CSCI-121 · Week 2, Day 2

Name _____ **Section (circle): R3 Mon / R4 Wed**

The rule from today

> **Casting truncates. Formatting rounds. Choose on purpose.**

```java
double v = 3.999;

(int) v                      // 3
String.format("%.0f", v)     // "4"
```

Strings — the set worth memorising

Java	Python
`s.length()`	`len(s)`
`s.charAt(i)`	`s[i]`
`s.substring(a, b)`	`s[a:b]` — stops **before** `b`
`s.equals(t)`	`s == t`
`s.toUpperCase()`	`s.upper()`
`s.trim()`	`s.strip()`
`s.contains(t)`	`t in s`

No negative indexing. The last character is `s.charAt(s.length() - 1)`.

Every one of these returns a new String. None of them change the one you had.

Predict, then we run it

	Code	Your prediction
1	`String s = "hi"; s.toUpperCase(); print(s);`	
2	`print("hello".substring(0, 3));`	
3	`print("hello".charAt(4));`	
4	`print((int) 2.99);`	
5	`print(String.format("%.0f", 2.99));`	
6	`print(((int)(1.4815 * 100)) / 100.0);`	
7	`print(((int)(1.4815 * 100)) / 100);`	

6 and 7 differ by one character. Say why:

The two-decimal trick — SB1 needs this

```
double v = 1.4815297665908702;
double t = ((int)(v * 100)) / 100.0;
```

Fill in each step:

Step	Expression	Value
1	`v * 100`	
2	`(int)(v * 100)`	
3	`... / 100.0`	

`printf` vs `String.format`

```
System.out.printf("%.2f%n", v);          // _____________________________
String out = String.format("%.2f", v);   // _____________________________
```

Which one would you use inside a method that has to **return** the text?

Compiling — what the terminal showed us

```
javac Hello.java
```

What new file appeared? _____

Is it for you to read? _____

The first four bytes of every Java class file ever compiled: _____

```
java Hello
```

Why is there no `.java` or `.class` on that command?

Compile time vs run time

	When it happens	Example
Compile time		
Run time		

`TimeUtils` — with your neighbour

3725 seconds. Fill in the blanks so this returns `01:02:05`.

```java
public static String formatTime(int totalSeconds) {

    int hours   = totalSeconds ______ 3600;

    int rem     = totalSeconds ______ 3600;

    int minutes = rem ______ 60;

    int seconds = rem ______ 60;

    return String.format("%02d:%02d:%02d", hours, minutes, seconds);
}
```

`3725 / 3600` is `1`, not `1.03`. Here integer division is not something you are working around — it is the tool. Why?

Exit ticket — hand this in at the door

1. `(int) 2.99` and `String.format("%.0f", 2.99)` — give both answers.

2. You run `javac` and get an error. How many `.class` files were produced?

3. One thing from Q1 you got wrong and now understand.

This week and next

Skill Builder 1	**Sun Sep 13, 11:59 PM** — today was its content
Drills — Asterisks · Show me the Numbers	**Sun Sep 6, 11:59 PM**
Recitation 2 — finish at home, push, open the PR	**Sun Sep 6, 11:59 PM**
Reading — Think Java **Ch 2**, **Ch 3**	
§6.10 *String Comparison* · **§9.3** *Strings Are Immutable*	the two sections that cover today and Tuesday

> **R3 — no section on Mon Sep 7. Labor Day.** Your next section is **Sep 14**, and SB1 is due Sep 13. Use office hours and tutoring, and re-read **Recitation 2 exercises 6, 7 and 8** — they are SB1's spine.